Amiga USB-C PD Power Supply v1.2 — Amiga 500 / 600 / 1200 (and other +5/+12/−12 V machines)
Needs a USB-C PD charger that offers 15 V at 3 A (45 W or more). U1 (CH224A) requests 15 V via R1 = 56k.
U2/U4 run only while the PD contract holds (PG low -> Q1 off) and VBUS > ~11.5 V (R3/R4).
+5 V: U2 SY8205 -> U3 eFuse (5.64 V OVLO, ~5.4 A limit); 4.5 A continuous. +12 V: U4, 12.36 V. −12 V: U5 inverting, from +12 V.
Budget: 45 W in, about 37 W out, e.g. 5 V 4.5 A + 12 V 1 A + −12 V 0.2 A.
Original design. Spec targets only from Commodore PSU ratings and the SnCMuseum V2 board; no circuitry copied.

PWR_FLAG PWR_FLAG PWR_FLAG PWR_FLAG PWR_FLAG
VBUS GND 5V_BUCK −12V +12V

J1 USB-C (power in) VBUS CC1 CC2 D− D− D+ D+ SBU1 SBU2 SHIELD GND SH A1 A4 A5 B5 A7 B7 A6 B6 A8 B8
U1 CH224A CC1 CC2 VBUS VDD PG CFG1 CFG2 CFG3 DM DP GND
C1 1uF/50V
R1 56k
U2 SY8205FCC IN IN EN BS LX FB VCC SS GND
C2 10uF/50V C3 10uF/50V C4 100nF/50V
C5 100nF/50V
L1 4.7uH
C10 100pF
C27 100nF/50V
C28 1uF/50V
R5 75k
R6 10k
U3 TPS25947OARPWR 5V_BUCK IN OUT AUXOFF EN/UVLO FLT OVLO ILM DVDT TIMER GND +5V EF_EN EF_OVLO EF_DVDT FLT EF_ILM
C6 22uF/25V C7 22uF/25V C8 22uF/25V C9 22uF/25V
R8 100k R10 100k R9 47k R11 27k
C11 2.2nF
R12 620R
D1 SMF5.0A C12 22uF/25V
R13 1k R14 1k
D3 RED D2 GREEN LED5
Q1 2N7002 PG EN
R2 100k R3 100k R4 12k
U4 LMR51430XDDCR VIN EN CB SW FB GND
C13 10uF/25V C14 10uF/25V C15 100nF/50V
C16 100nF/50V
L2 12uH
R15 100k R16 5.1k
C17 22uF/25V C18 22uF/25V C19 22uF/25V
D4 SMF13A
R17 4.7k
D5 GREEN LED12
U5 LMR51430XDDCR VIN EN CB SW FB GND
C20 10uF/50V C21 10uF/50V C22 100nF/50V
C23 100nF/50V
L3 12uH
R18 100k R19 20k R20 100k R21 5.1k R22 4.7k
D8 GREEN LEDN
C24 22uF/25V C25 22uF/25V C26 22uF/25V
D6 B5819W D7 SMF13A
J2 AMIGA OUT −12V GND +5V +5V GND +12V

USB-C PD 15 V -> Amiga +5 V 4.5 A (eFuse protected), +12 V, −12 V

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File: Amiga USB-C PD Power Supply.kicad_sch		
Title: Amiga USB-C PD Power Supply		
Size: A3	Date: 2026-11-03	**Rev: 1.2**
KiCad E.D.A. 10.0.4		Id: 1/1